business rules implemented on the server may not need to exercise the UI at all. In the rest of this chapter we'll discuss ways to effectively automate acceptance criteria like these without the need for automated UI tests.

9.3 Types of non-UI automated acceptance tests

As you've seen, UI tests have their place when it comes to testing and illustrating user interactions with the system and the user experience as a whole, but it's often inefficient to use them for more detailed business rules. There are a number of strategies that can be used to bypass heavyweight UI tests when automating your acceptance criteria, and many tools can be used to implement these approaches. Some of the more commonly used approaches include

- Testing against the controller-layer application code
- Testing directly against business logic in the application code
- Testing services remotely, such as by invoking web services and thus avoiding the UI layer entirely

Figure 9.4 illustrates these strategies in use with a typical modern web application.

In the following sections, we'll look at each of these options in more detail.

9.3.1 Testing against the controller layer

In some cases, you may be able to implement workflow and validation acceptance criteria without touching the web interface at all. This is particularly true if your application uses one of the many variations of the Model-View-Controller (MVC) architecture. MVC is a widely used architecture pattern that aims at cleanly separating the data an application uses from the way it's presented to the user (see figure 9.5).

If your application uses a variation of the MVC architecture, you may be able to write some of your acceptance tests directly against the controller layer. This is sometimes

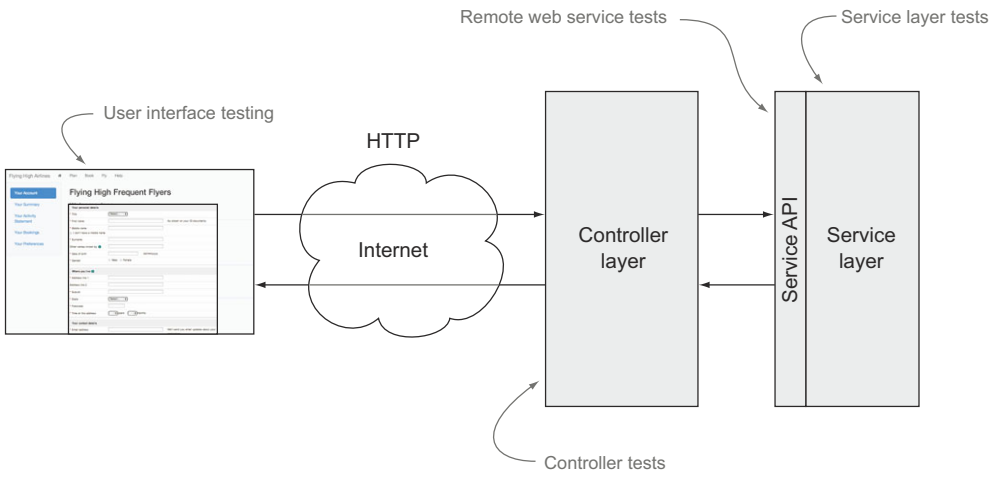


Figure 9.4 Applications need different types of tests for different parts of the system.